

TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 1

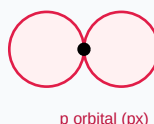
CHAPTER 1: BASIC CONCEPTS OF CHEMISTRY AND CALCULATIONS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Basic Concepts of Chemistry and Calculations

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

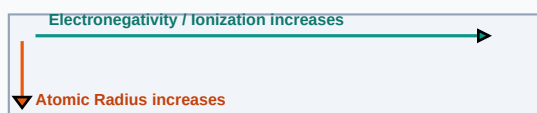
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Mole	The SI unit of amount of substance containing exactly 6.022×10^{23} elementary entities.
Equivalent Mass	The mass of a substance which combines with or displaces 1g of hydrogen, 8g of oxygen, or 35.5g of chlorine.
Limiting Reagent	The reactant in a chemical reaction that is completely consumed first, limiting the amount of product formed.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

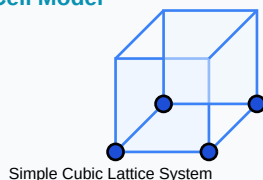
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Number of Moles: $\text{Moles} = \text{Given Mass} / \text{Molar Mass}$
- Gram Equivalent: $\text{Eq} = \text{Molar Mass} / \text{Equivalence Factor (Acidity / Basicity)}$
- Empirical Formula: Simplest ratio of constituent atoms in a compound

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

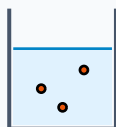
Example 1: Calculate equivalent mass of Sulphuric Acid (H_2SO_4 , Molar Mass = 98 g/mol).

Step-by-step Solution:

H_2SO_4 is dibasic, so equivalence factor = 2.

Equivalent Mass = Molar Mass / Basicity = $98 / 2 = 49 \text{ g/eq}$.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: If 50kg of Nitrogen gas and 10kg of Hydrogen gas are mixed to produce Ammonia (NH₃), identify the limiting reagent.

Analytical Solution Strategy:

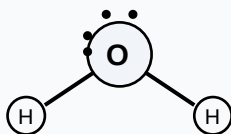
Reaction: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$.

Moles $\text{N}_2 = 50000 / 28 = 1785.7$ moles | Moles $\text{H}_2 = 10000 / 2 = 5000$ moles.

Required H_2 for $\text{N}_2 = 1785.7 \times 3 = 5357$ moles.

Since only 5000 moles H_2 are available, Hydrogen is the limiting reagent.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

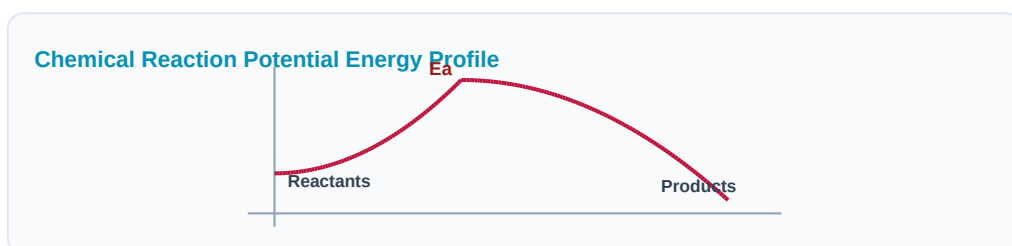
Q1 (1-Mark): Define relative atomic mass.

Q2 (2-Mark): Calculate molar mass of Glucose ($C_6H_{12}O_6$).

Q3 (2-Mark): Explain equivalent mass of an oxidizing agent with an example.

Q4 (5-Mark): Determine empirical formula of a compound having 40% C, 6.7% H, and 53.3% O.

Q5 (5-Mark): Differentiate between empirical formula and molecular formula.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Pharmaceuticals

Precise mole ratios ensure drug tablets contain correct stoichiometric values of active ingredients.

Application Case: Metallurgy

Furnace yields are optimized by calculating limiting ore inputs to prevent fuel waste.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

